Understanding the median

Task A: Finding the middle in small data sets

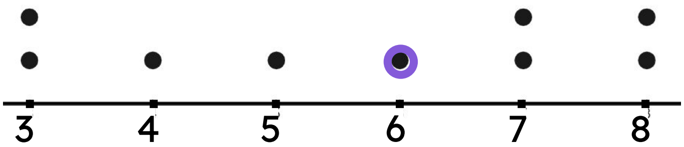
1) Each tray has some cupcakes.
What is the **median** number of cupcakes?

4 cupcakes

2 *3* *4* *5* *6*

2) Some data are shown on the dot plot.
What is the **median** value?

6



3) If possible, find the **median** for these data sets

a) 16, 32, 29, 19, 30, 42, 51, 29, 25, 28

16, 19, 25, 28, 29, 29 30, 32, 42, 51
*the **median** is 29*

b) avocado, beetroot, carrot, damson, edamame, fig

*no **median** for these data*

c) -3, -9, -3, -7, -2, -10, -1

-10, -9, -7, -3, -3, -2, -1
*the **median** is -3*

Task B: Finding the middle in large data sets

1) Work out the position of the **median** in each case

a) 425 data points — *213^{th} piece of data*

b) 327 data points — *164^{th} piece of data*

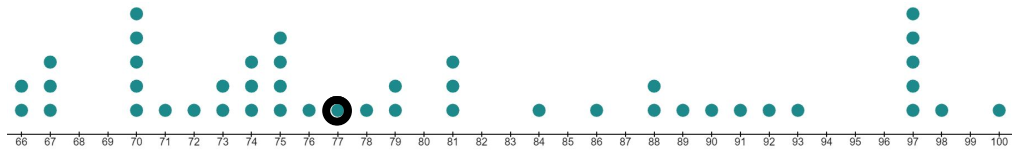
c) 2389 data points — *1195^{th} piece of data*

d) 412 data points — *206.5^{th} piece of data*
- the mean of the 206^{th} and 207^{th} data values

e) 5686 data points — *2843.5^{th} piece of data*
- the mean of the 2843^{rd} and 2844^{th} data values

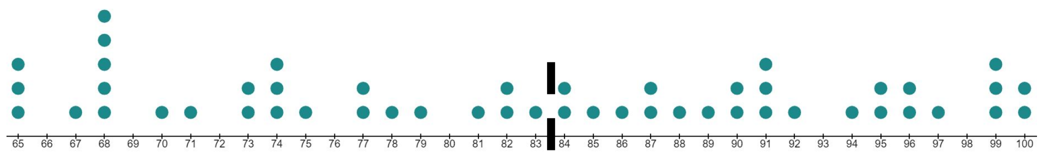
2) Work out the median from each dot plot

a)



77

b)



83.5

3) Here there are 97 pieces of data in order.

4.4, 4.6, 5.4, 5.5, 5.7, 6, 6.2, 6.5, 6.6, 6.8, 7.5, 7.9, 8.2, 8.3, 8.4, 8.4, 8.8, 8.8, 8.8, 9.2, 9.5, 9.6, 9.6, 9.6, 9.8, 9.9, 9.9, 10.1, 10.2, 10.3, 10.3, 10.5, 10.6, 10.9, 10.9, 11.2, 11.3, 11.4, 11.6, 11.6, 11.8, 12.3, 12.5, 12.6, 13.2, 13.3, 13.3, 13.9, 13.9, 14.2, 14.2, 14.4, 14.5, 14.8, 15.3, 15.4, 15.4, 15.5, 16.1, 16.6, 16.6, 17.1, 17.3, 17.3, 17.4, 17.7, 17.7, 17.8, 18.2, 18.3, 18.3, 18.3, 18.4, 18.4, 18.5, 18.5, 18.7, 18.9, 18.9, 19, 19.1, 19.2, 19.3, 19.6, 20, 20, 20.1, 20.9, 20.9, 21.1, 21.2, 21.6, 22.1, 23.1, 23.3, 23.5, 25.9

What is the **median** value? *The piece of data in the 49^{th} position → 13.9*

4) There are 2000 items in the data set in total. The part of the list shown starts at the 998th entry. What is the **median** for this data set?

…, −4.5, −4.5, −4.2, −3.9, −3.3, −2.1, −2.0, −0.8, −0.1, 0, 0, 0.2, 0.2, 0.2, 1.7, 1.8, …

998th

***Median** = −4.05*